David Hoffman

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SUMMARY

Engineer with 15+ years of experience building production-grade software systems for network automation, cybersecurity, sports technology, and electronics R&D. Educated thousands of engineers through written technical content including blogs, books, and courses. Recently completed formal quantum networking and quantum computing coursework through University of Chicago and Chicago Quantum Exchange with hands-on simulation of quantum circuits and quantum network protocols. Focused on applying a broad systems engineering background across software, hardware, and networking, along with clear communication style to emerging quantum technologies.

TECHNICAL SKILLS

Programming: Python, C, SQL, JavaScript, Bash, React

Frameworks & APIs: Django, Flask, GraphQL, REST APIs, WebSockets

Networking & Security: Nautobot, Ansible, Itential, Keycloak, Vault

AI & Computer Vision: TensorFlow, PyTorch, mxnet OpenCV, Scikit-image, Scikit-learn, Copilot, Claude, Agentic-systems

Data & Databases: PostgreSQL, MySQL, Neo4J, Redis, Google Datastore/Firestore, Hadoop

Cloud & Infrastructure: AWS (Lambda, S3, EC2), GCP (Compute, Run, Kubernetes, Storage), Render.com

DevOps: Docker, CI/CD, Git, Jenkins

QUANTUM COMPUTING & QUANTUM NETWORKING EXPERIENCE

- Simulated quantum algorithms and communication protocols (Grover's, Simon's, superdense coding, teleportation, BB84 QKD) using Qiskit with Python/Jupyter, progressing from simple gates and EPR pairs to multi-step systems
- Demonstrated a star-topology quantum network in Python using SeQUeNCe, configuring hardware and protocol parameters to model entanglement requests between nodes, and analyzed performance metrics across varying network conditions

PROFESSIONAL EXPERIENCE

Vice President, Software Engineer II, Network Automation — Bank of America, Core Technology Infrastructure

June 2025 - present – Chicago, IL

- Enhance global network automation platform technology and operations across Americas, Europe, and Asia-Pacific
- Integrated Nautobot SSoT (single source of truth) to Itential device sync for over 140K network devices globally
- Resolved over 15+ after-hours critical platform incidents and followed up with root cause analysis
- Executed 8+ successful software releases and trained overseas team to handle and own releases going forward via documentation and knowledge transfer session delivery
- Organized Confluence documentation repository dating back to 2019 according to the PARA (Projects, Areas, Resources, Archive) method so that documentation producers and consumers can easily create and digest information

Founder & Software Engineer — BocceLabs LLC, privately held firm serving sports communities and leagues

January 2023 - present – Chicago, IL (part-time)

- Built a sports technology software platform from the ground up providing bocce leagues with live data services
- Lead four-person team across engineering, product, and operations
- Architected system delivering 100+ matches per season (Ohio Bocce League, Chicago Bocce League, Chicago Bocce Club) with automated scoring and stats overlays for the Bocce Broadcast Network YouTube channel
- Developed real-time data pipelines to/from cloud services (Django, Redis, Postgres) and multiple on-prem devices (tablets, smartphones, digital scoreboards)
- Implemented cloud infrastructure with 0-downtime CI/CD enabling reliable deployments and rapid iteration
- Scaled platform to 200+ league users across three states with recurring seasonal engagement

Senior Software Engineer, Cybersecurity — Reveald Inc., privately held cybersecurity firm serving municipalities

June 2022 - May 2025 – Remote (Chicago, IL)

- Lead backend developer for cybersecurity vulnerability, exposure, impact, and agent control
- Developed graph reduction algorithms using NetworkX and custom methods to simplify node-edge relationships, enabling efficient browser rendering and improved user visualization

- Created a cross-platform ticket management integration for JIRA and ServiceNow saving $150K in annual license fees
- Implemented a system health backend and dashboard for system troubleshooting and status saving 200h annual troubleshooting time

Senior Software Engineer, Document Management Systems — Deloitte via DivIHN Integration consulting group

March 2021 - June 2022 – Remote (Chicago, IL)

- Architected a high-volume automated document management system generating $10M ARR
- Integrated document classification and text recognition (OCR) engines (DocumentAI, HyperScience, Parascript)
- Certified SCRUM Master & SCRUM Product Owner

Senior Software Engineer, Writer, & **Customer Support Engineer** — PyImageSearch, privately held independent blog

May 2017 - March 2021 – Remote (Durham, NC & Chicago, IL)

- Developed advanced computer vision and AI pipelines for image classification, object detection, tracking and segmentation
- Authored 170+ technical tutorials and a 3-vol book raising $370K on Kickstarter
- Educated and supported 15,000+ engineer customers globally translating complex concepts into applied solutions
- Scaled technical audience from from 80,000 to 300,000

Software Engineer, Network Test Automation — Cisco Systems via GDH Consulting

June 2017 - October 2017 – Raleigh/Durham, NC

- Implemented Python network automation test cases (BGP, Segment Routing) for Cisco customers
- Configured switches, routers, and traffic generators with Python automation

Graduate Research Assistant — East Carolina University, Computer Science Department

May 2016 - May 2017 – Greenville, NC

- Instructor for 60-student undergraduate Discrete Mathematics course delivering 2x lectures and 1x office hour weekly
- Awarded the 2016-2017 Outstanding Computer Science Graduate Assistant by the department
- Designed Hadoop research cluster for handwriting recognition research

Electrical Engineer, High Performance Cable Test Systems — Molex, High Performance Cables

May 2014 - May 2016 – Little Rock, AR

- Developed and maintained automated test systems for high-speed electrical connectors in R&D labs
- Created Python, TCL, and C scripts to interface with oscilloscopes, spectrum analyzers, TDRs, and VNAs, ensuring high-frequency signal accuracy for datacenter cables
- Analyzed cable and connector signal integrity and built automated reporting tools to streamline data analysis and visualization for engineering and quality teams
- Supported global manufacturing sites for quality control and troubleshooting (Little Rock, Guadalajara, Dongguan, Manila)

RF Engineer, Cellular, Microwave, & Radar — Northrop Grumman, Naval Surface Warfare Center

August 2011 - April 2014 – Morehead City, NC (MCAS Cherry Point) & Virginia Beach, VA (NAS Oceana)

- Designed, integrated, tested, and secured communication systems for Navy/Marine Corps tactical training ranges
- Performed RF signal analysis and link budgets using MATLAB RF Toolbox and custom spreadsheet models and troubleshooted wireless systems using spectrum and network analyzers
- Installed antennas, lightning protection, and VSWR equipment on east coast radio towers (rescue-climbing certified)
- Procured and tracked hardware/software for DoD projects up to $300K in asset inventory systems
- Trained technicians and military personnel to operate and maintain radar and wireless equipment

EDUCATION

Certificate – **Quantum Science, Networking, and Communications** — University of Chicago – 2025

Masters of Science – **Software Engineering** — East Carolina University – 2017

Bachelors of Science – **Computer Engineering** — North Carolina State University – 2011

Bachelors of Science – **Electrical Engineering** — North Carolina State University – 2011